

Robust Optimization: Assignment 1

Davide Carecci¹, Benjamin Riedel²

¹Davide Carecci is with the Department of Electronics, Information and Bioengineering, Politecnico di Milano

²Benjamin Riedel is with the Department of Business Intelligence, Friedrich Schiller Universität Jena

I. UNCERTAINTY SET DESIGN

Visit the link of COLAB for Exercises 3.1, 3.2 and 3.3: Exercises 3.1-3

A. Exercise 3.1

Γ around 3 allows 3 out of 10 coordinates to take extreme (vertexes) in one scenario. Coherently with geometric considerations, Γ must be higher than γ to contain the same number of scenario (95%).

B. Exercise 3.2

$\Gamma_{CC} \geq \Gamma$ and $\gamma_{CC} \geq \gamma$ were obtained. Since the standardized real data distribution already satisfies **Assumption 3.4** of the lecture notes, this is coherent with the statements of **Theorem 3.5** and **Corollary 3.8**, that holds for any distribution under the above mentioned assumptions. Real data distribution are thus characterized by a lower variance and higher lower bounds with respect to other possible symmetric independent distributions defined over $[-1,1]$ (Fig. 1).

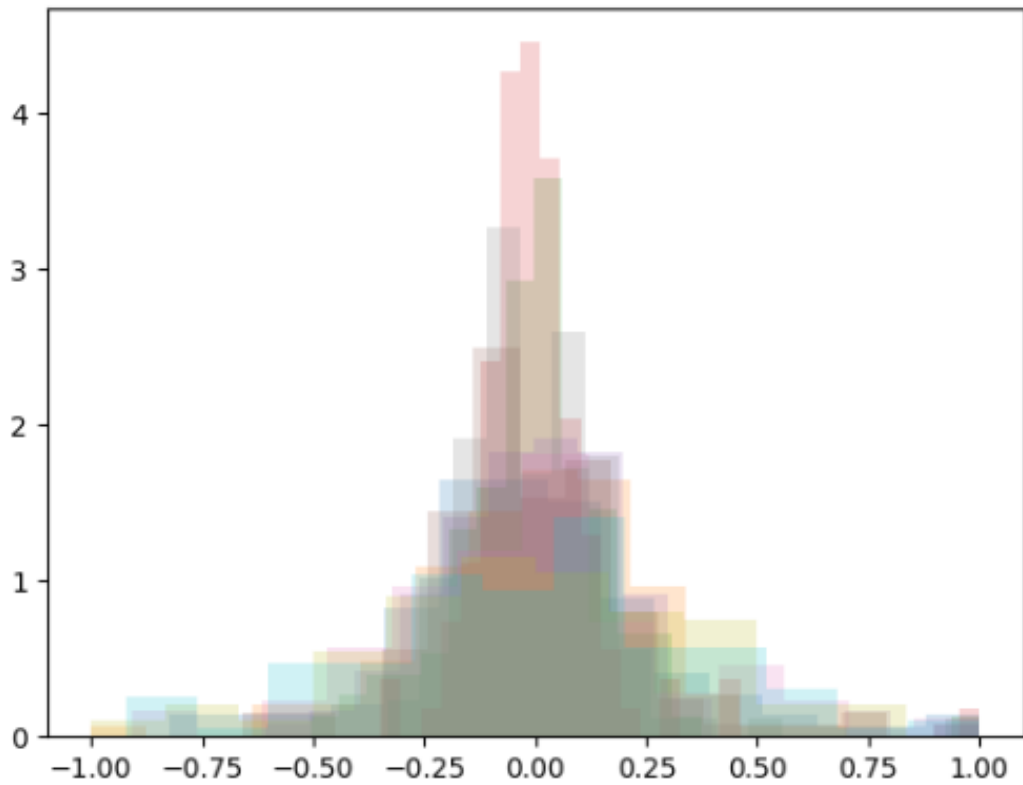


Fig. 1. Histograms of "normalized" data. One column for each item.

C. Exercise 3.3

$VaR|_{ell}$ from Theorem 3.5 is more conservative than $VaR|_{ell}$ calculated from real calibration and validation data. Nevertheless, extreme distribution scenario are not "covered by the calibrated γ ". Similar considerations hold for the estimated $VaR|_{bud}$. Even though $\Gamma \geq \gamma$, $VaR|_{ell}$ is more conservative than $VaR|_{bud}$.

Estimated $VaR|_{CVaR}$ is even less conservative, but, on the other hand, much more coherent with the $VaR|_{CVaR}$ obtained on real data. It is risky to optimize based on this set, since $VaR|_{CVaR}$ is lower than the VaR value obtained over the 2010-2014 scenarios. There are not enough scenarios (Datapoints) to get a generalizing $CVaR$ -Set.

$VaR|_{ell}$ and $VaR|_{bud}$ from **Theorem 3.5** and **Theorem 3.8** are equal when optimizing over the designed sets in **Exercise 3.2** and both grasp the worst-case scenarios characterized by very low probability. This is suggested since both values are equal to the VaR calculated over the extreme distribution of data. Coherently, when increasing γ_{CC} or Γ_{CC} , the same $VaR|$ value is obtained: a lower bound with 100% confidence is found as far as the returns can be described by symmetric and i.i.d.'s distributions (original starting assumption). It was observed that, in this case, x^* contains all zeros but for one item, that was found to be the one characterized by the scenario vector with the highest lower bound (i.e. the safest one). The z^* of this worst case scenario are coherently all zero but -1 for the that item entry.

II. REFORMULATION FOR RISK AVERSE PRODUCTION PROBLEM: EXERCISE 3.4

Visit the link of COLAB for Exercises 3.4: Exercises 3.4

A. Reformulation of agent A balance constraint

The relation between z and Δ specified in the description of $\mathcal{Z}$ was used to rewrite the original constraint as a function of Δ . The constraint was rewritten as $\bar{x}^T \Delta \leq -d^T y + \hat{z}^T x$, $\forall z \in \mathcal{Z}$ where $\hat{x} = -\hat{z}x$. From the description of the set $\mathcal{Z}$, 3 constraints were derived and consequently 3 additional variables were defined:

- 1) $\lambda^+ \in \mathbb{R}^n$ associated to $\Delta_i \leq 1 \forall i = 1..n$
- 2) $\lambda^- \in \mathbb{R}^n$ associated to $-\Delta_i \leq 1 \forall i = 1..n$
- 3) $\gamma \in \mathbb{R}^1$ associated to $-\sum_{i=1}^n \Delta_i \leq \Gamma \sqrt{n}$

The W matrix was obtained as:

$$W = \begin{bmatrix} \mathbb{I} \\ -\mathbb{I} \\ -1^T \end{bmatrix}$$

The v vector was obtained as:

$$v = \begin{bmatrix} 1 \\ 1 \\ \Gamma \sqrt{n} \end{bmatrix}$$

B. Reformulation of objective function upper-bound constraint

The constraint was rewritten as $\bar{y}^T q \leq -s - c^T x$, $\forall q \in \mathcal{Q}$ where $\bar{y}^T = -P^T y$. From the description of the set $\mathcal{Q}$, 4 constraints were derived and consequently 4 additional variables were defined:

- 1) $\gamma^+ \in \mathbb{R}^1$ associated to $-\sum_{i=1}^N q_i \leq 1$
- 2) $\gamma^- \in \mathbb{R}^1$ associated to $\sum_{i=1}^N q_i \leq 1$
- 3) $\lambda^+ \in \mathbb{R}^N$ associated to $q_i \leq \gamma \bar{q}_i \forall i = 1..N$
- 4) $\lambda^- \in \mathbb{R}^N$ associated to $-q_i \leq -\gamma^{-1} \bar{q}_i \forall i = 1..N$

The W matrix was obtained as:

$$W = \begin{bmatrix} -1^T \\ 1^T \\ \mathbb{I} \\ -\mathbb{I} \end{bmatrix}$$

The v vector was obtained as:

$$v = \begin{bmatrix} -1 \\ 1 \\ \gamma \bar{q} \\ -\gamma^{-1} \bar{q} \end{bmatrix}$$

With the help of LP-Duality and the reformulation technique used in class we obtained the "Reduced RiskAverse Production Problem".